

NATIONAL INSTITUTE OF TECHNOLOGY ROURKELA

■ Database Engineering CS2008 ■ End Sem Spring 2023-24 ■ UG ■ Pg 1 ■ FM 50 ■ 3 Hrs

■ Answer ANY TWO from Question# 1, 2, 3, and Answer BOTH Question# 4, 5. ■

■ All parts of a question MUST be answered together. ■ Mere answers without proper explanation will not fetch marks. ■ Variation in quality of answer will vary the secured marks. ■

1. ~~a.~~ What is the criteria for second normal form?
~~b.~~ Consider the relation scheme $R = \{E, F, G, H, I, J, K, L, M\}$ and the given set of functional dependencies as $\{ \{E, F\} \rightarrow \{G\}, \{F\} \rightarrow \{I, J\}, \{E, H\} \rightarrow \{K, L\}, \{K\} \rightarrow \{M\}, \{L\} \rightarrow \{N\} \}$ on R . What is the key for R ?
~~c.~~ Explain what is a Surrogate key with an example.
~~d.~~ What is view of a table and when it is used?
~~e.~~ Explain what is denormalization with an example. [2 × 5]

2. ~~a.~~ What is a schedule? Check if the following schedule S on the three transactions T_1, T_2, T_3 are (i) view equivalent or not, (ii) conflict equivalent or not, (iii) recoverable schedule or not, (iv) cascading schedule or a cascadeless schedule. $S: \{R_1(A), W_2(A), W_1(A), R_2(A), W_3(A), Commit_2, Commit_1, Commit_3\}$.
~~b.~~ Represent division with basic relational algebra operators.
~~c.~~ How minimum participation is related to existence dependency? Explain with one example bearing two entities and one relationship. [(1+4)+2+3]

3. ~~a.~~ Check with suitable algorithm if $BD \rightarrow E, AC \rightarrow E$ individually are the redundant FDs in $F = \{A \rightarrow B, C \rightarrow D, BD \rightarrow E, AC \rightarrow E\}$.
~~b.~~ What do you mean by soundness and completeness of Armstrong's axioms?
~~c.~~ Given a relation $R(A, B, C)$ with functional dependency set as $F = \{A \rightarrow B, B \rightarrow C\}$, find F^+ . *well*
~~d.~~ Draw the symbol in ER diagram to represent a weak entity and an identifying relationship.
~~e.~~ If a ternary relationship connecting three entities (with $(0, N)$ participation from two entities and $(0, 1)$ participation from one entity) is converted to a relation in relationship schema, minimum how many relations will be required to represent the entities and the relationship, and why? How many foreign keys will be required in total, and why? [3 + 1 + 2.5 + 1 + 2.5]

4. ~~a.~~ State and Present the difference between the Standard subquery and correlated subquery and Join with an example.
~~b.~~ For a schema $R(A, B, C, D, E, F, G, H)$ bearing functional dependency $F = \{AB \rightarrow CEFGH, A \rightarrow D, F \rightarrow G, FB \rightarrow H, HBC \rightarrow ADEFG, FBC \rightarrow ADE\}$, check if F is in BCNF or not? And if not, convert it into BCNF.
~~c.~~ For a schema $R(A, B, C, D, E, F)$ having FD $G = \{A \rightarrow B, A \rightarrow C, C \rightarrow D, AE \rightarrow F\}$, find if the decomposition of R into $R_1(A, B, C), R_2(C, D)$, and $R_3(A, E, F)$ is lossless.
~~d.~~ For a schema, $R(A, B, C, D)$ bearing FD $F = \{A \rightarrow B, B \rightarrow C, C \rightarrow D, D \rightarrow A\}$, find if the decomposition of R into $R_1(A, B, C)$ and $R_2(B, C, D)$ is dependency preserving?
~~e.~~ When will the output of SELECT in SQL and π in relational algebra be same? Can they be different in any case? Explain both scenarios with examples. [3+3+3+3+3]

5. ~~a.~~ What is alias in SQL and when is it used?
~~b.~~ Find the content of tables $G, H, I, J, K, L, M, N, P$ from the following tables R and S :
 $G \leftarrow R \bowtie_{A=D} S \quad H \leftarrow R \ltimes_{B=E} S \quad I \leftarrow R \ltimes_{C=D} S \quad J \leftarrow R \ltimes_{A>D} S \quad K \leftarrow R * S$
 $L \leftarrow R \triangleright_{A=F} S \quad M \leftarrow R \ltimes_{A<D} S \quad N \leftarrow R \ltimes_{B>E} S \quad P \leftarrow R \ltimes_{A=E} S$ [1.5 + 1.5 × 9]

R(A, B, C)			S(D, E, F)		
A	B	C	D	E	F
5	6	3	2	8	5
4	1	4	9	5	8
8	5	3	1	7	4
2	3	9	4	7	9
5	7	6	6	3	1

■ BEST OF LUCK ■